

AWS Academy Machine Learning Foundations

Module 5: Introducing Computer Vision

Sections

1. Introducing computer vision
2. Analyzing images and videos
3. Preparing custom datasets for computer vision
4. Module wrap-up

Demonstrations

1. Introducing Amazon Rekognition
2. Labeling Images with Amazon SageMaker Ground Truth

Lab

Guided Lab: Facial Recognition



**Knowledge
check**

At the end of this module, you should be able to:

- Describe the use cases for computer vision
- Describe the Amazon managed machine learning (ML) services for image and video analysis
- List the steps required to prepare a custom dataset for object detection
- Describe how Amazon SageMaker Ground Truth can be used to prepare a custom dataset
- Use Amazon Rekognition to perform facial detection

Module 5: Introducing Computer Vision

Section 1: Introducing computer vision

Computer vision is the automated extraction of information from digital images.

Computer vision applications



Public safety
and home security



Authentication and enhanced
computer-human interaction



Content management
and analysis



Autonomous driving



Medical imaging



Manufacturing process control

Computer vision problems

Content recognition

Image analysis

- Object classification



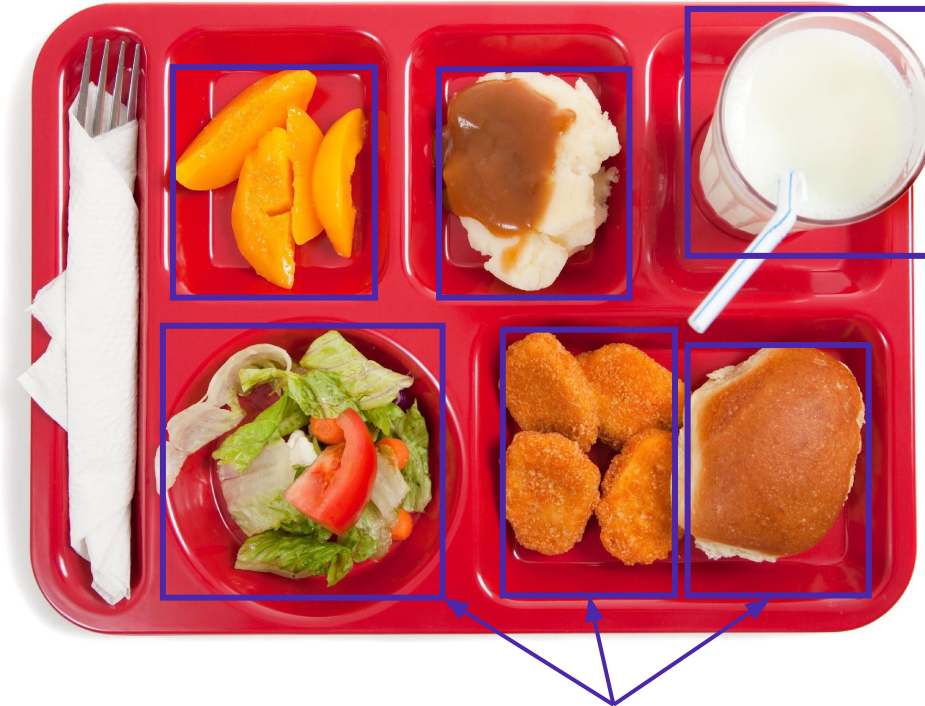
Food?
Breakfast
?
Lunch?
Dinner?

Computer vision problems

Content recognition

Image analysis

- Object classification
- Object detection



Bounding
boxes

(top,left,width,height)

Confidence
e

Milk	97.1
Peaches	92.3
Ice Cream	97.1
Salad	69.5
Nuggets	77.5
Bread Roll	94.5

Computer vision problems

Content recognition

Image analysis

- Object classification
- Object detection
- Object segmentation



Milk

Peaches

Ice Cream

Salad

Nuggets

Bread Roll

Content recognition

Video analysis

- Instance tracking

Pathing – You can capture the path of people in the scene. For example, you can use the movement of athletes during a game to identify plays for post-game analysis.

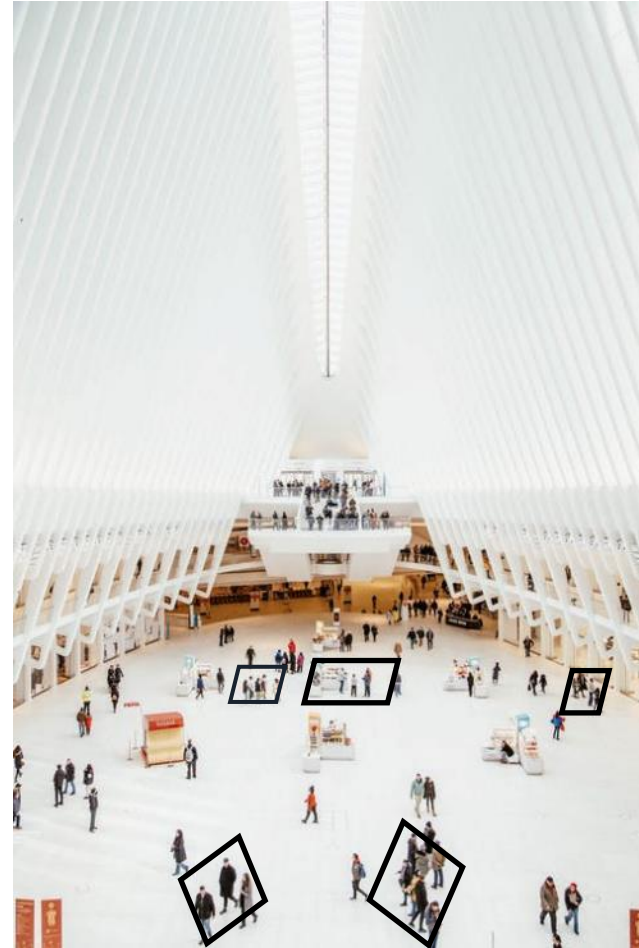


Content recognition

Video analysis

- Instance tracking
- Action recognition

Analyze shopper behavior and density in your retail store by studying the path that each person follows



Content recognition

Video analysis

- Instance tracking
- Action recognition
- Motion estimation



Section 1 key takeaways



- Computer vision is the automated extraction of information from images
- Image analysis includes object classification, detection, and segmentation
- Video analysis includes instance tracking, action recognition, and motion estimation

Module 5: Introducing Computer Vision

Section 2: Analyzing images and videos

Amazon Rekognition

- Managed service for image and video analysis
- Types of analysis
 - Searchable image and video libraries
 - Face-based user verification
 - Sentiment and demographic analysis
 - Unsafe content detection
 - Text detection
- Security and compliance



Amazon Rekognition



Can add powerful visual analysis to your application



Is highly scalable and continuously learns



Integrates with other AWS services

Languages supported by the Amazon Rekognition SDKs:

JavaScript

Python

PHP

.NET

Ruby

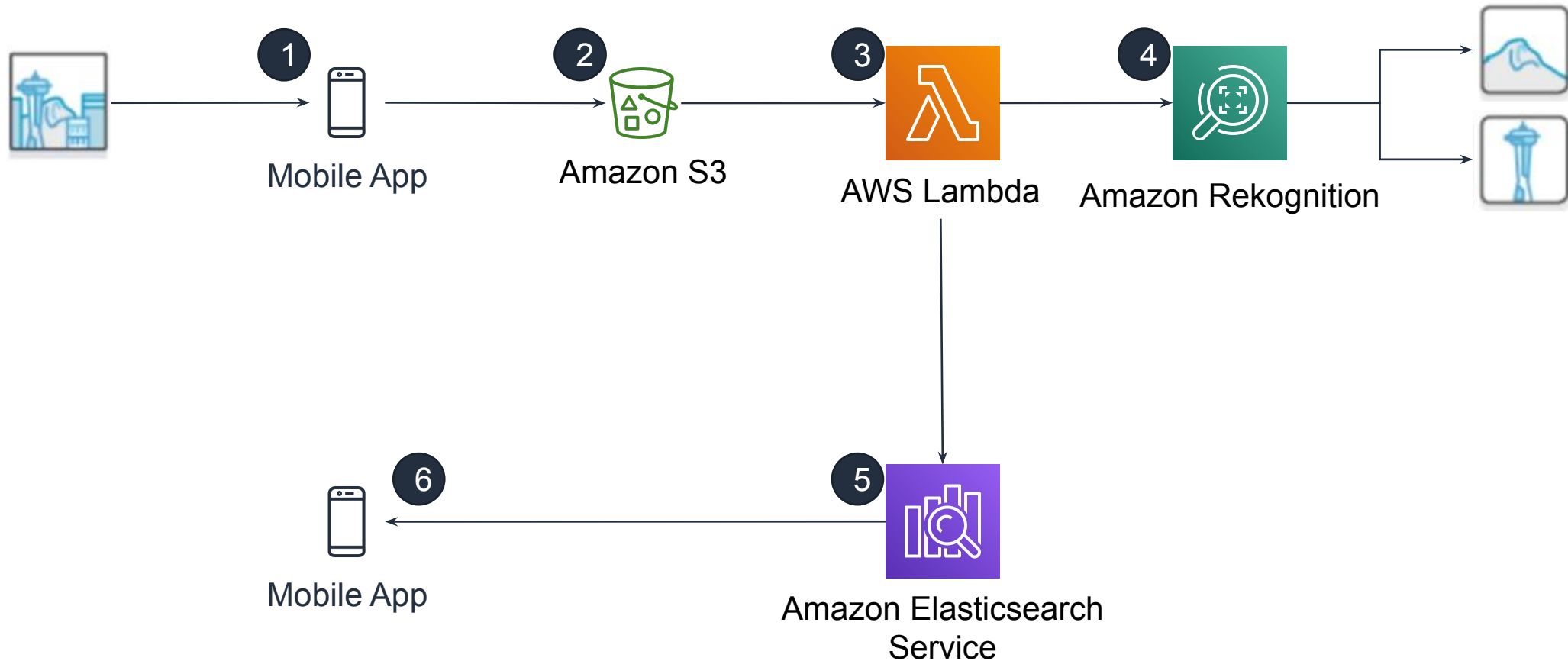
Java

Go

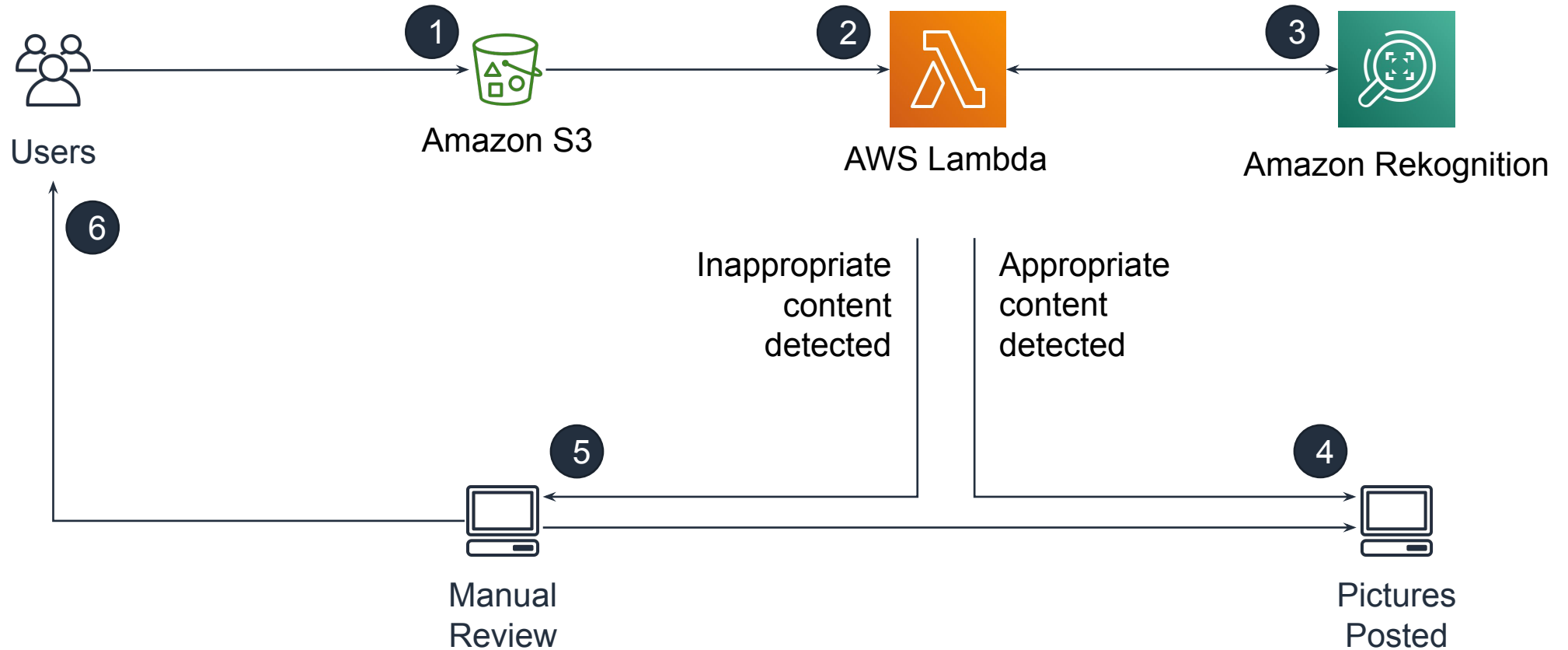
Node.js

C++

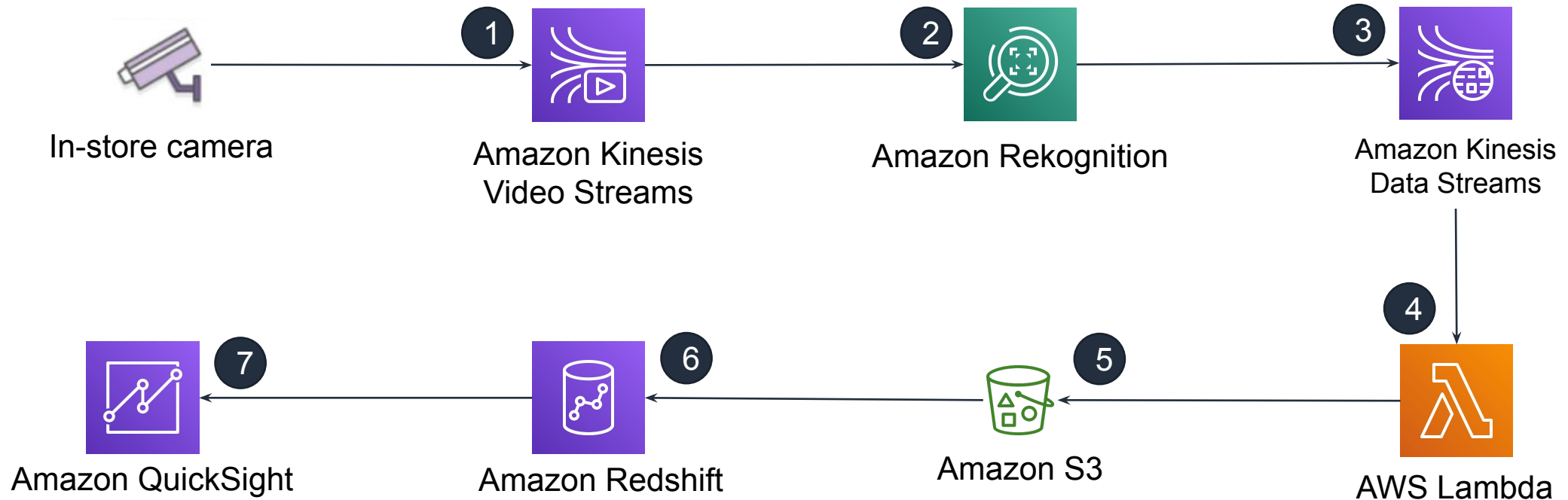
Use case 1: Searchable image library



Use case 2: Image moderation



Use case 3: Sentiment analysis



Using Amazon Rekognition

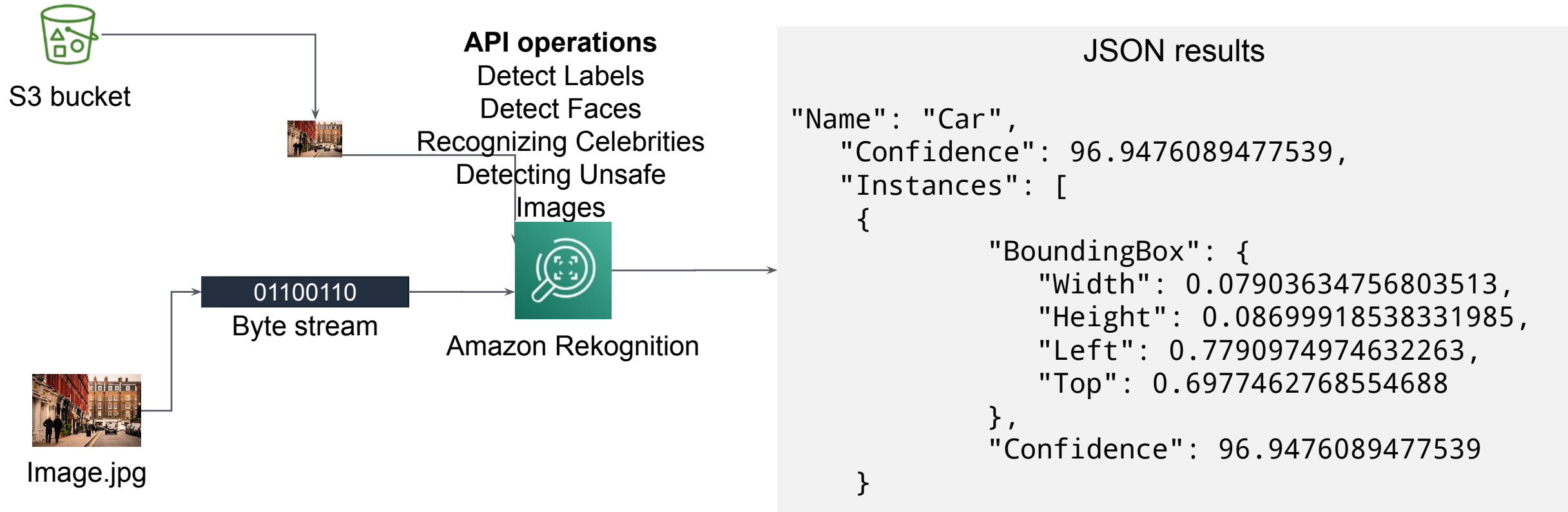


Image analysis: Object and scene detection

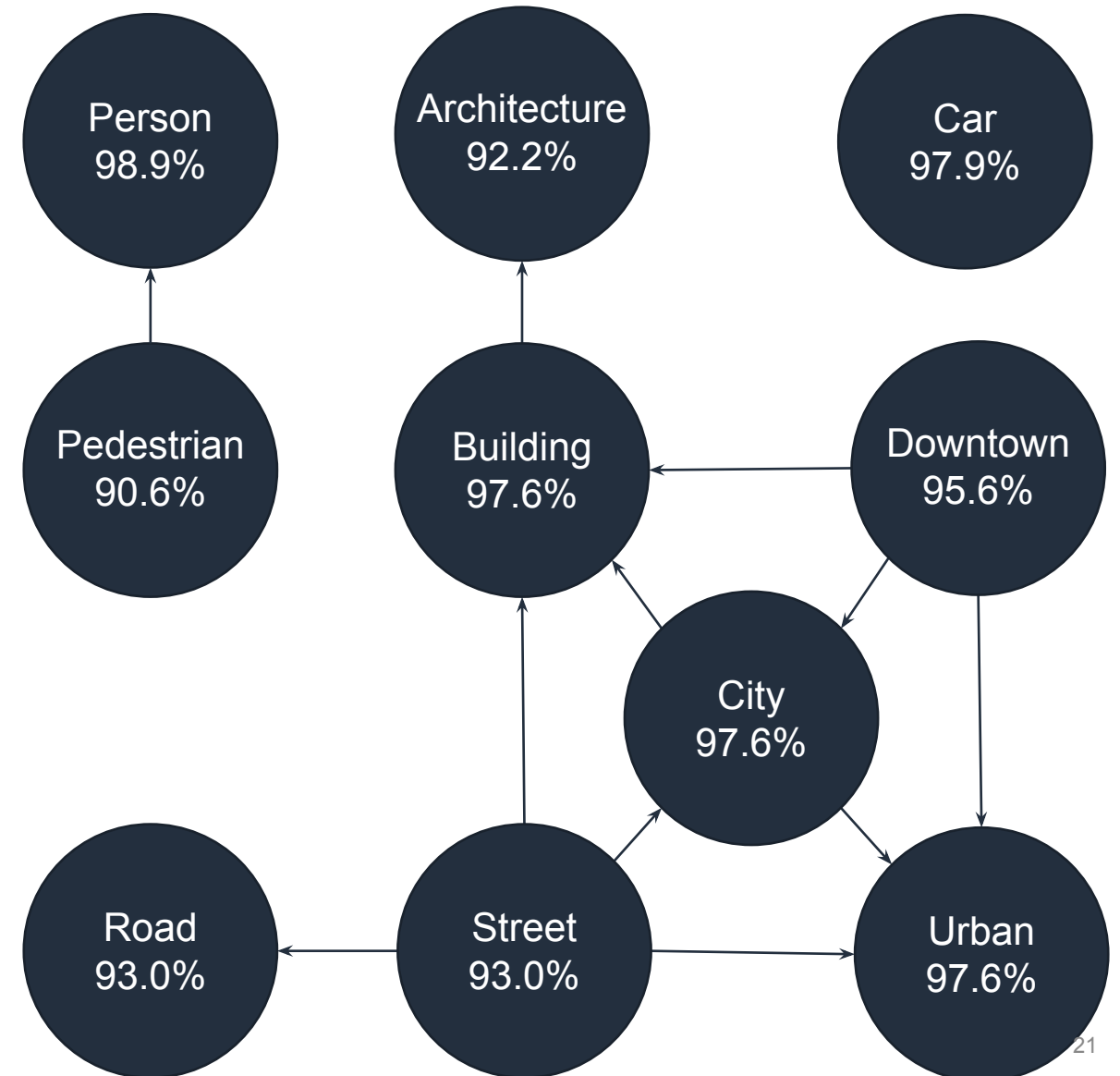
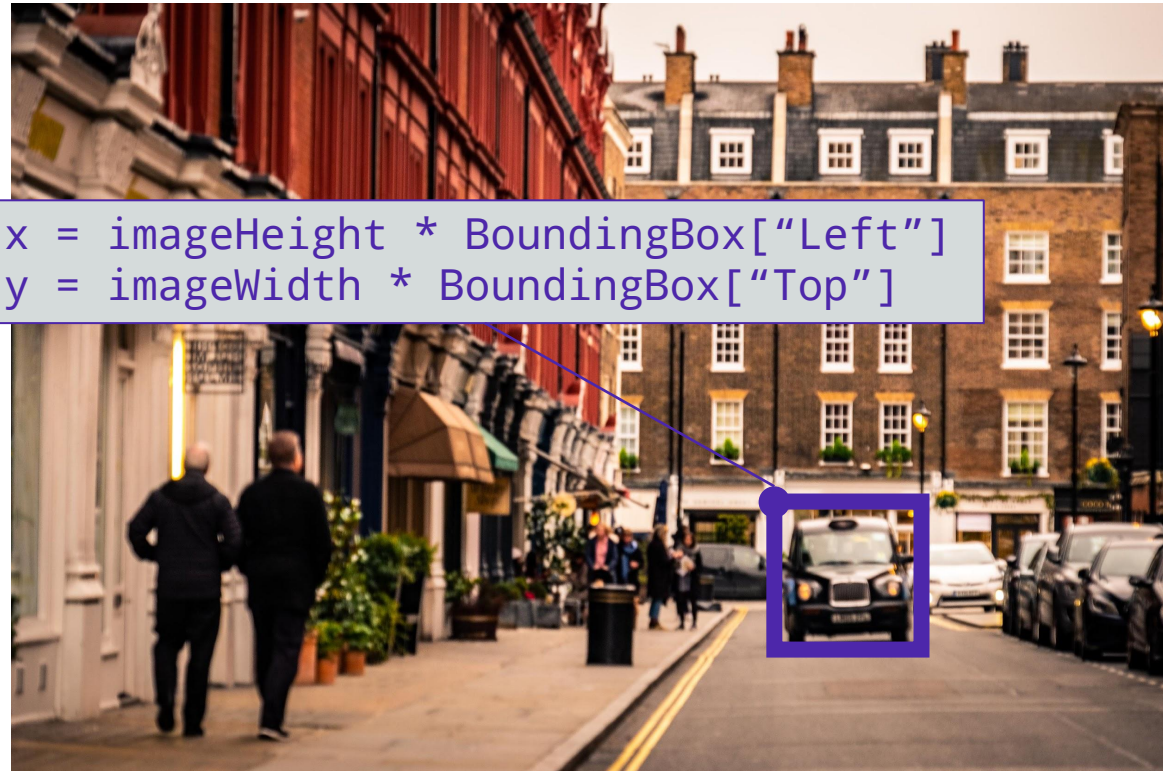


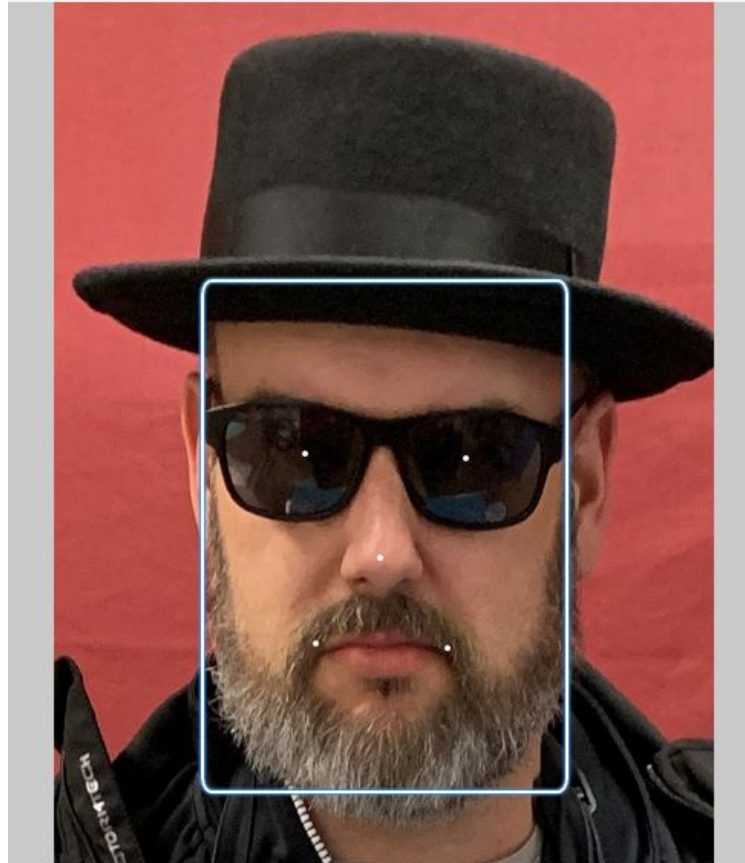
Image analysis: Object and scene detection



```
"Name": "Car",  
  "Confidence": 96.9476089477539,  
  "Instances": [  
    {  
      "BoundingBox": {  
        "Width": 0.07903634756803513,  
        "Height": 0.08699918538331985,  
        "Left": 0.7790974974632263,  
        "Top": 0.6977462768554688  
      },  
      "Confidence": 96.9476089477539  
    }  
  ]  
}
```


Face detection

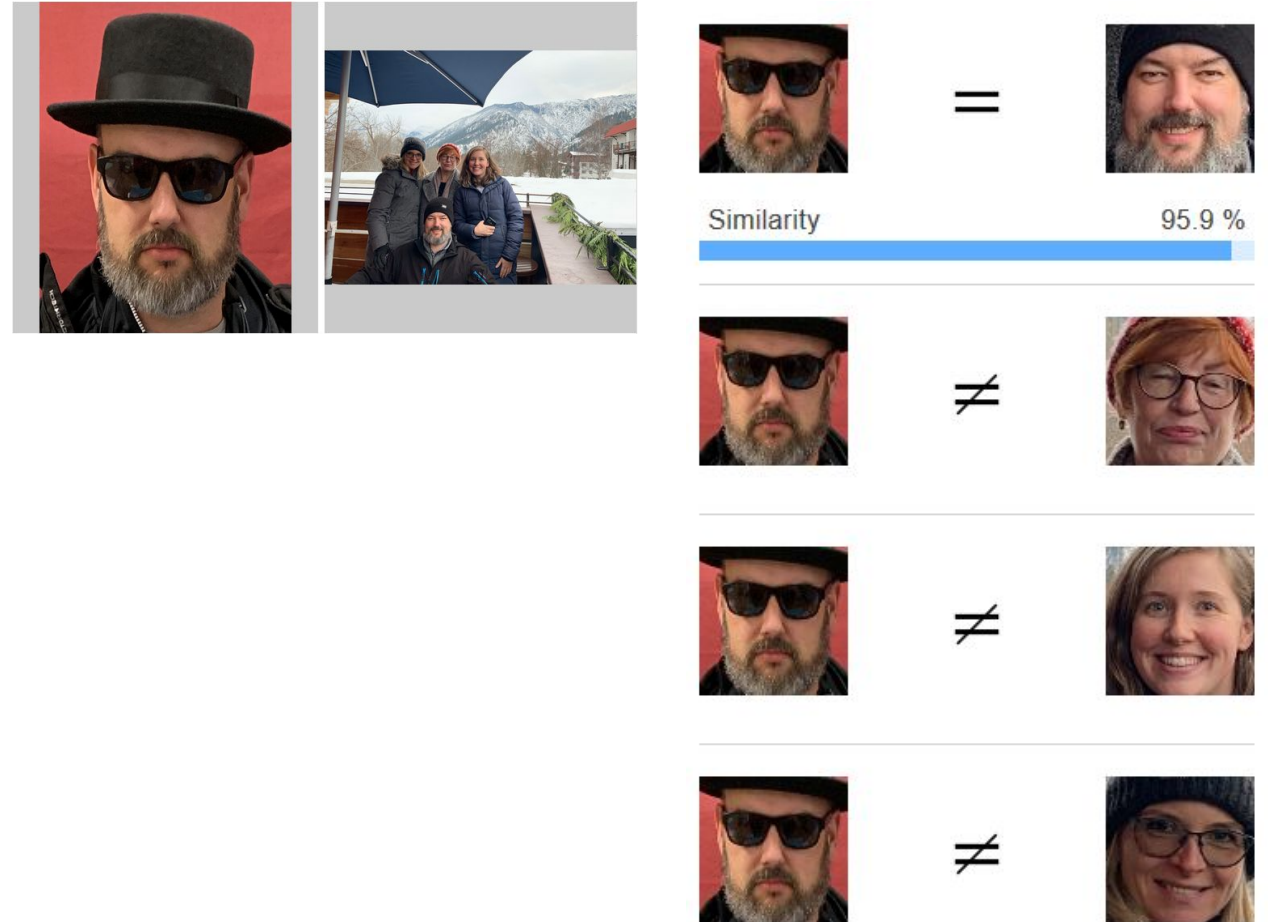
- Bounding box
- Attributes
- Emotions
- Facial landmarks
- Quality
- Pose
- Confidence score



looks like a face	99.9 %
appears to be male	99.5 %
age range	32 - 48 years old
not smiling	99.8 %
appears to be calm	99.4 %
wearing glasses	99.5 %
wearing sunglasses	96.7 %
eyes are open	99.9 %
mouth is closed	98.7 %
does not have a mustache	75.1 %
has a beard	96.6 %

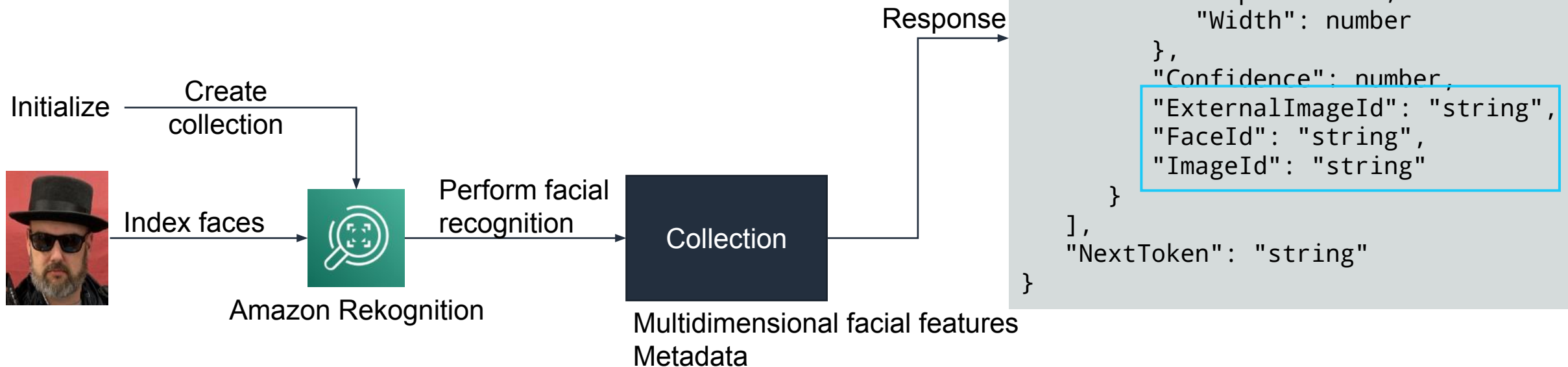
Using facial recognition

- Compare *source* with *target*
- Face match –
 - Bounding box and confidence
 - Similarity score
 - Facial landmark locations
- Source face information –
 - Bounding box and confidence
 - Facial landmarks
- Unmatched face –
 - Bounding box and confidence
 - Facial landmarks



Searching for known faces

- Must train the model
- Use images with target faces
- Perform facial recognition
- Store facial metadata



Searching for Known Faces



SearchFacesByImage



Amazon Rekognition

Collection

Multidimensional facial features
Metadata

Response

```
{ "FaceMatches": [ {  
  "Face": {  
    "BoundingBox": { "Height": 0.0633333027, "Left": 0.171851992, "Top": 0.73666697,  
"Width": 0.1106169968 },  
    "Confidence": 100,  
    "ExternalImageId": "input.jpg",  
    "FaceId": "578e2e1b-d0b0-493c-aa39-ba476a421a34",  
    "ImageId": "9ba38e68-35b6-5509-9d2e-fcffa75d1653" },  
    "Similarity": 99.9764175415039 } ],  
  "FaceModelVersion": "3.0",  
  "SearchedFaceBoundingBox": { "Height": 0.063333332, "Left": 0.171851858, "Top":  
0.73666667, "Width": 0.11061728 },  
  "SearchedFaceConfidence": 99.99999237060547 }
```

- Facial detection
 - Bounding box, attributes, emotions, landmarks, quality, pose
 - Confidence score
- Detection is based on image data
 - Gender based on image, not identity
 - Emotions are inferred from physical appearance
- You can use the appropriate confidence score for your use case
- Facial recognition should never be used in a way that violates an individual's rights, including the right to privacy, or makes autonomous decisions for scenarios that require analysis by a human*

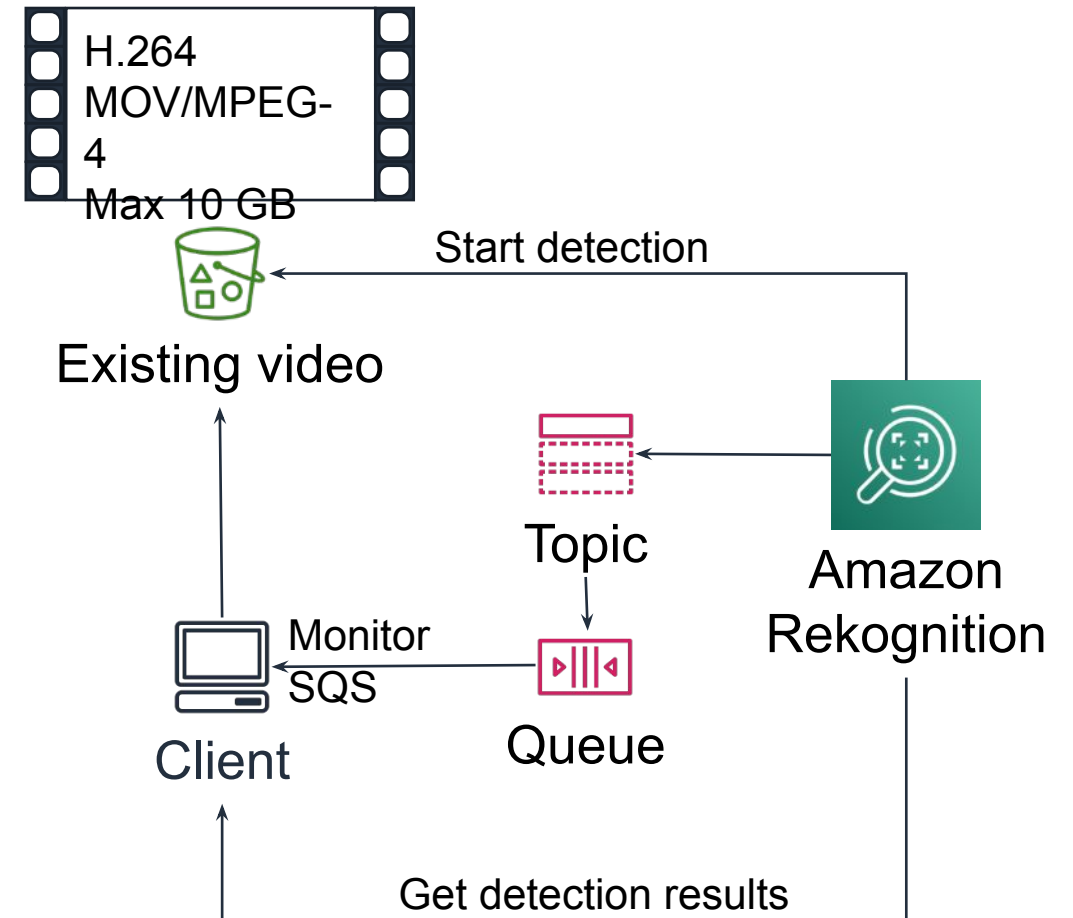
Demonstration: Introducing Amazon Rekognition



Working with stored videos

Process:

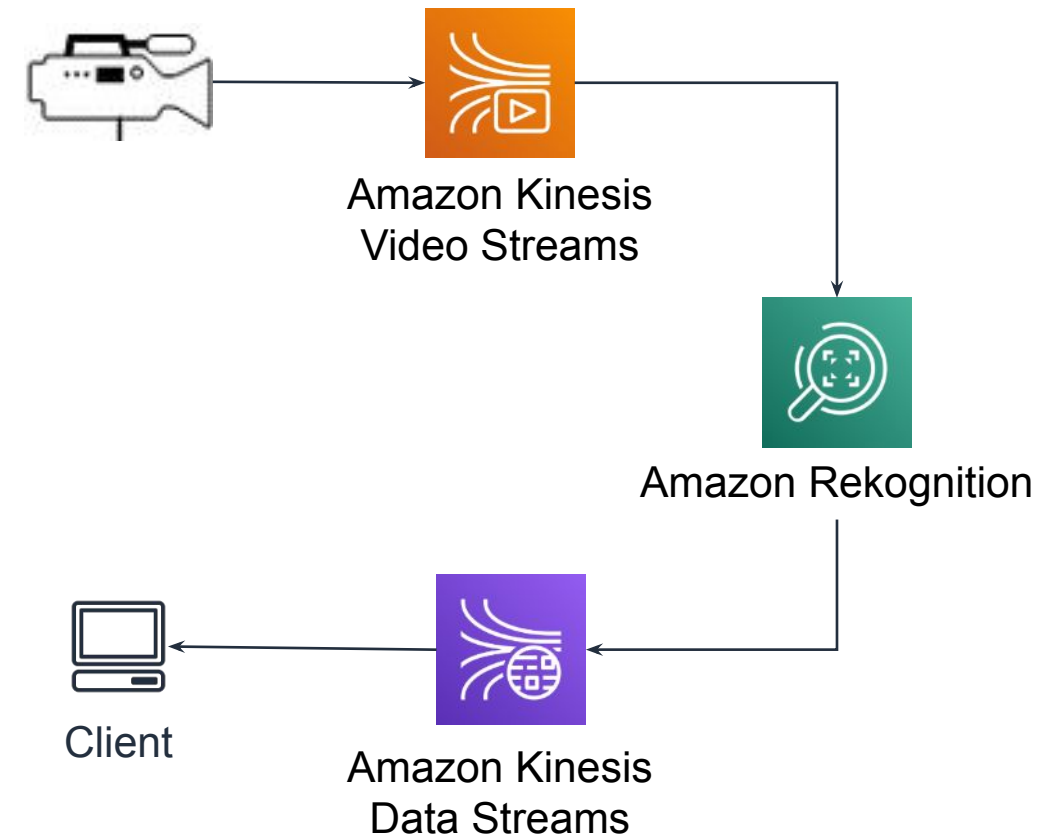
- Start detection
 - People, faces, labels, celebrities, text, inappropriate content
- Monitor Amazon Simple Queue Service (Amazon SQS) queue for completion
- Get detection results



Working with streaming videos

Application process:

1. Stream video to Amazon Kinesis Video Streams
2. Connect Amazon Rekognition Video stream processor
3. Read analysis from the Amazon Kinesis data stream



Section 2 Key Takeaways



- You can use Amazon Rekognition for image and video analysis that uses proven, highly scalable, deep learning technology that doesn't require machine learning expertise
- Provides image and video detection of faces, sentiment, text, unsafe content, and library search
- Is integrated into other AWS services

Module 5: Introducing Computer Vision

Section 3: Preparing custom datasets for computer vision

AWS DeepLens Powered Cat Flap



Models need training on domain

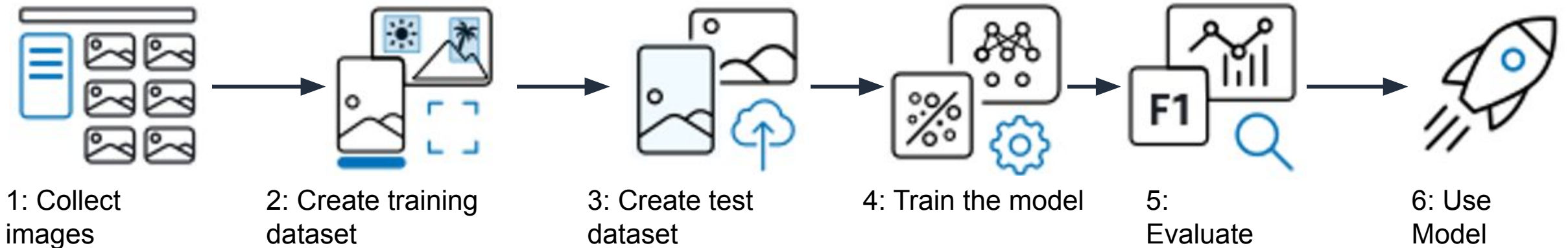


Wood	97%
Canvas	88.7%
Text	84.7%
Envelope	71.7%
Greeting Card	70.4%
Mail	70.4%
Hardwood	60.6%
Plywood	60.4%
Paper	57.7%
Advertisement	55.8%
Poster	55.8%
Art	55.6%

Amazon Rekognition Custom Labels

- Use for –
 - Search for logos
 - Identify products
 - Identify machine parts
 - Distinguish between healthy and infected plants
- Almost all vision solutions start with an existing model
- Benefits –
 - Simplified data labeling
 - Automated machine learning
 - Simplified model evaluation, inference, and feedback

Custom labeling process



Step 1: Collect images



- Typically use a few hundred images
- Build domain-specific models
- Use 10 PNG or JPEG images per label
- Use images similar to the images that you want to detect

Step 2: Create training dataset



- Dataset: Data about images, labels, and bounding box
- Create at least two labels
- Label the images by using the console or Amazon SageMaker Ground Truth

Image-level versus object-level labels

Scenes and concepts



Label:
beach

Objects with bounding boxes



Label: *Echo
Dot*

Images need labels



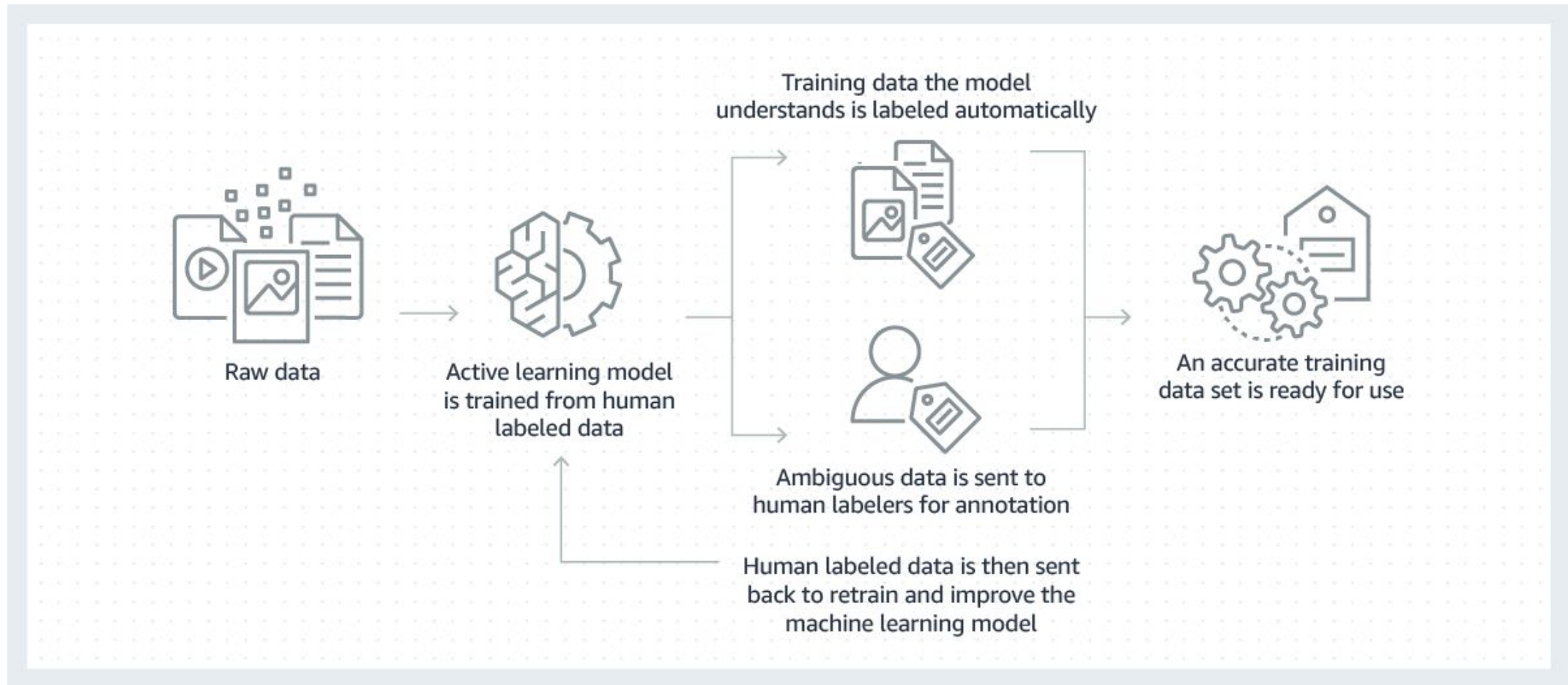
```
{  
  "source-ref": "s3://b/.../img_2783.jpg",  
  "small_beach": 1,  
  "small_beach-metadata": {  
    "confidence": 1,  
    "job-name": "labeling-job/small_beach",  
    "class-name": "Beach",  
    "human-annotated": "yes",  
    "creation-date": "2020-03-02T20:42:03.525Z",  
    "type": "groundtruth/image-classification"  
  }  
}
```


Objects need bounding boxes

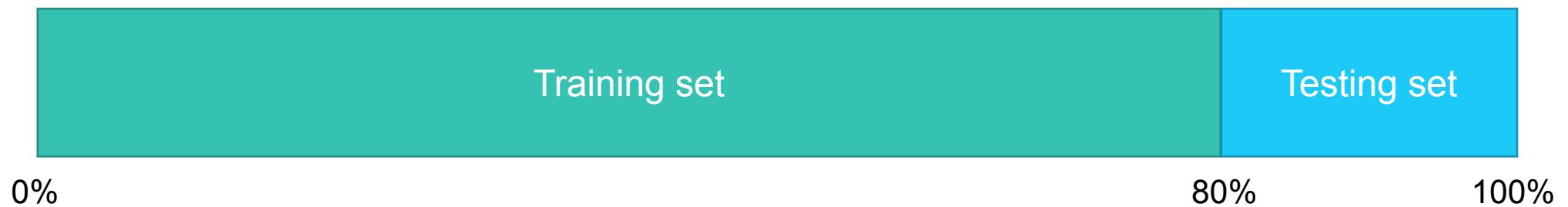
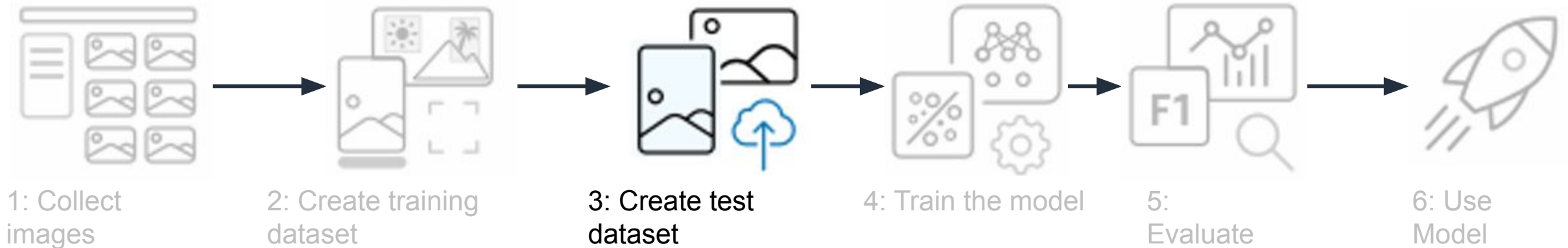


```
"small_BB": { "annotations": [
  { "left": 186,"top": 125,"width": 346,"height": 580,
    "annotationType": "bounding-box", "id": 5,"class_id": 0},
  { "left": 732,"top": 445,"width": 270,"height": 192,
    "annotationType": "bounding-box", "id": 6,"class_id": 1}],
  "image_size": [
    { "width": 1152,"height": 864,"depth": 3 }]],
  "small_BB-metadata": {
    "job-name": "labeling-job/small_BB",
    "class-map": { "0": "Echo", "1": "Echo dot" },
    "human-annotated": "yes",
    "objects": [ {"confidence": 1}, {"confidence": 1}],
    "creation-date": "2020-03-02T20:42:25.930Z",
    "type": "groundtruth/object-detection"
```

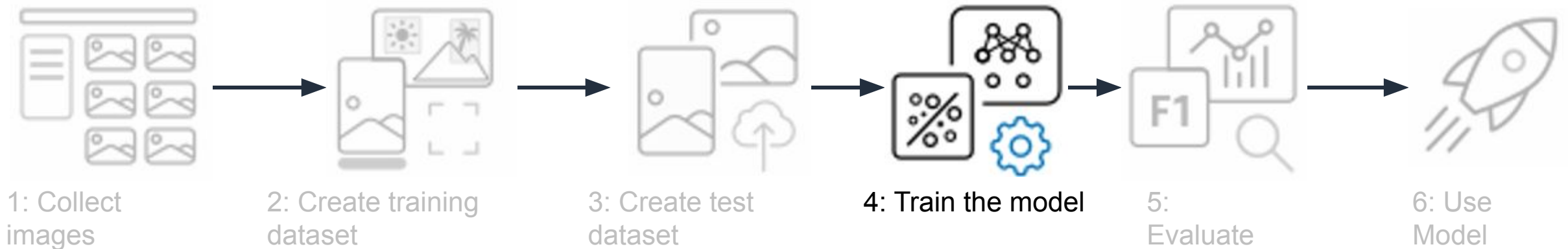
Amazon SageMaker Ground Truth



Step 3: Create test dataset



Step 4: Train the model



Step 5: Evaluate - Metrics



- Evaluate model performance

- Metrics

- Precision
- Recall
- Overall model performance

Confusion Matrix

		Actual	
		Cat	Not a Cat
P r e d i c t	Cat	TP	FP
	Not a Cat	FN	TN

Step 5: Evaluate – Improve



- Data
- Reducing false positives (better precision)
 - Adjust the confidence threshold to improve precision
 - Add additional classes as labels for training
- Reducing false negatives (better recall)
 - Lower the confidence threshold to improve recall
 - Use better data or more precise classes (labels) for training

Step 6: Use the model



```
aws rekognition detect-custom-labels --project-version-arn "model_arn"\
  --image '{"S3Object":{"Bucket":"bucket","Name":"image"}}'\
  --min-confidence 70
```

Returns array of custom labels:

- Label
- Bounding box for objects
- Confidence

Demonstration: Labeling Images with Amazon SageMaker Ground Truth

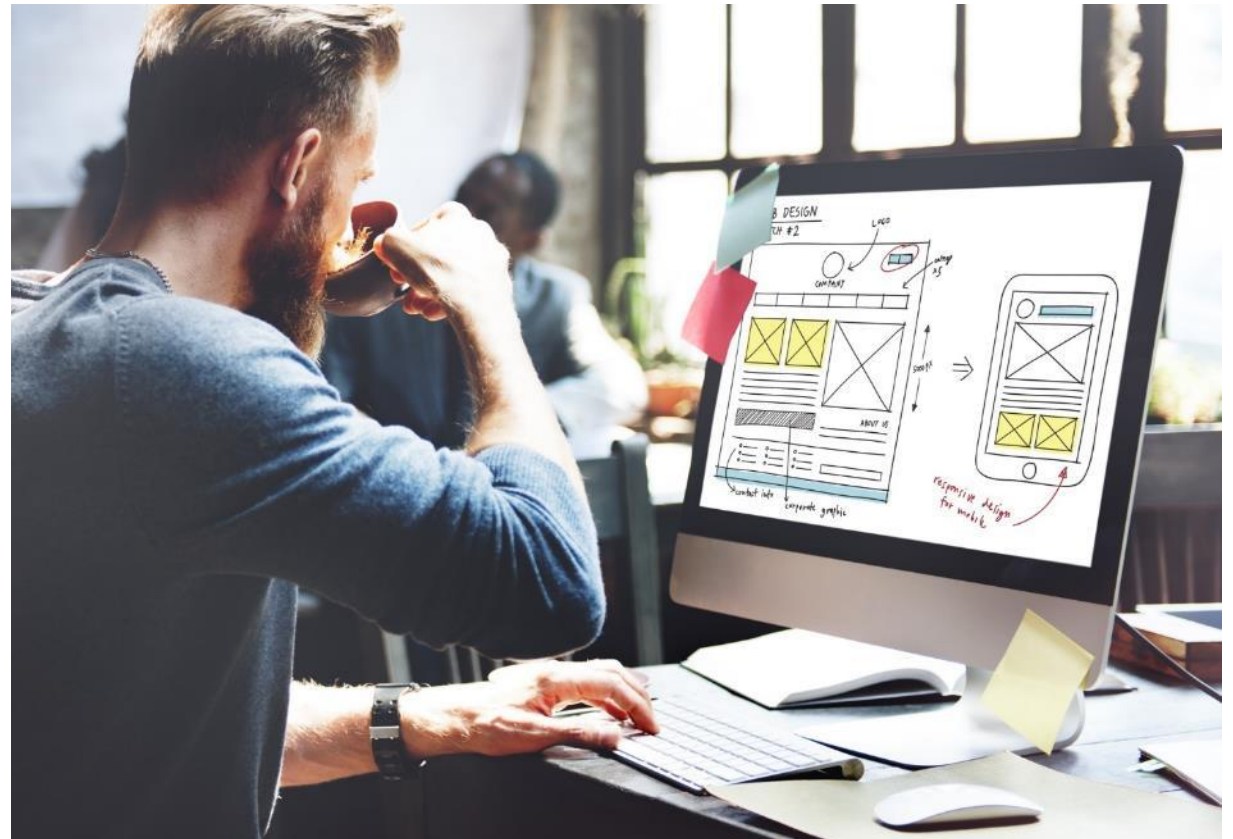


Section 3 key takeaways



- Models must be trained for the specific domain that you want to analyze
- You can set custom labeling for the specific business case
 - Custom labeling workflow
- You must label images and create bounding boxes for objects
- You can use Amazon SageMaker Ground Truth to build training datasets for your models

Module 5 – Guided Lab: Facial Recognition



Module 5: Introducing Computer Vision

Module wrap-up

In summary, in this module, you learned how to:

- Describe use cases for computer vision
- Describe the AWS managed machine learning (ML) services for image and video analysis
- List the steps required to prepare a custom dataset for object detection
- Describe how Amazon SageMaker Ground Truth can be used to prepare a custom dataset
- Use Amazon Rekognition to perform facial detection

Complete the knowledge check



- [What is Amazon Rekognition](#)
- [Welcoming Amazon Rekognition Video: Deep-Learning Based Video Recognition](#)
- [Classify a Large Number of Images with Amazon Rekognition and AWS Batch](#)

Thank you